

Pre-Flight API Input Verification

Systematic validation of data integrity before S2 LOOCV

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Overview

Before running expensive S2 LOOCV API calls, this notebook runs 12 systematic checks on the data pipeline outputs. Each test produces a PASS/WARN/FAIL result. All tests must pass (or have documented exceptions) before proceeding.

Test 1: Chunking Completeness

Every document in the corpus should be fully chunked: the first chunk starts at page 1 and the last chunk ends at or beyond the document's page count.

	used	(Mb)	gc trigger	(Mb)	max used	(Mb)
Ncells	1982546	105.9	3882704	207.4	3882704	207.4
Vcells	74173091	565.9	128830636	982.9	124988368	953.6

Test 1: PASS

Test 2: No Short Chunks (min_chars Filter)

`make_chunks()` drops chunks with 100 or fewer characters as extraction artifacts (`min_chars = 100L`). This test verifies the filter is active.

Test 2: PASS (min_chars filter active)

Test 3: No Text-Less Tier Rows

Every Tier 1 and Tier 2 row must have non-empty text after the join from `assemble_c1_chunk_data()`.

Test 3: PASS

Test 4: Act Name Alignment

The act names in `aligned_data` should match exactly with the act names in the tier data (both directions).

Test 4: PASS

Test 5: Passage Delimiter Integrity

For acts with multiple passages (`n_passages > 1`), splitting on `\n\n` should produce multiple non-empty elements.

Test 5: PASS

Test 6: Usable Passages Per Act

After splitting on `\n\n` and filtering to passages with more than 50 characters, every act should have at least one usable passage.

Test 6: PASS

Test 7: No Within-Tier Duplicates

Tier 1 should have no duplicate (`doc_id`, `chunk_id`, `act_name`, `passage_idx`) combinations. Tier 2 should have no duplicate (`doc_id`, `chunk_id`, `act_name`) combinations. Multiple passages matching the same chunk are expected for Tier 1 and flagged as WARN.

Test 7: PASS

Test 8: Token Budget Fit

All chunks plus overhead (codebook prompt + few-shot examples + system prompt + output buffer) must fit within the 200K context window.

Component	Tokens (est.)
Codebook prompt	1,622
Few-shot examples	3,100
System prompt	125
Output buffer	2,000
Max chunk	61,058
Total	67,905

Test 8: PASS (max 67,905 tokens vs. 200,000 limit)

Test 9: Text Encoding Safety

Sample chunks from each tier and check for null bytes and control characters that could corrupt API calls.

Tier	Sampled	Unsafe
Tier 1	70	0
Tier 2	100	0
Negative	100	0

Test 9: PASS

Test 10: Year Filter Consistency

The tier data should respect the `max_doc_year = 2007` parameter. No chunk in Tier 1 or Tier 2 should come from a document published after 2007.

Test 10: PASS

Test 11: Few-Shot Example Preview

Generate examples for one LOOCV fold (first act held out) and verify they render correctly. This confirms that `generate_c1_loocv_fold_examples()` produces usable training data.

Held-out act: Changes in Depreciation Guidelines and Revenue Act of 1962

Examples generated: 10 (5 positive, 5 negative)

Example 1: FISCAL_MEASURE

Act: Tax Reform Act of 1969

Reasoning: This passage describes the Tax Reform Act of 1969, an enacted fiscal measure with substantive detail about its provisions.

Input (truncated):

The tax bill passed in December reduced revenues for the next fiscal year by close to \$3 billion, compared to my original proposals

Example 2: FISCAL_MEASURE

Act: Revenue Act of 1948

Reasoning: This passage describes the Revenue Act of 1948, an enacted fiscal measure with substantive detail about its provisions.

Input (truncated):

should business begin to taper off, the effects of the tax reduction upon incentives to work and to invest would be extremely important, as would also the additional purchasing power generated by the increase in exemptions

Example 3: FISCAL_MEASURE

Act: Tax Reform Act of 1969

Reasoning: This passage describes the Tax Reform Act of 1969, an enacted fiscal measure with substantive detail about its provisions.

Input (truncated):

Total output would have to rise significantly faster than the growth rate of potential output

Example 4: FISCAL_MEASURE**Act:** Tax Adjustment Act of 1966**Reasoning:** This passage describes the Tax Adjustment Act of 1966, an enacted fiscal measure with substantive detail about its provisions.**Input (truncated):**

The combined effect of budgeted expenditures ... and tax laws now in effect would be more stimulative than now seems appropriate for the period ahead

Example 5: FISCAL_MEASURE**Act:** Economic Recovery Tax Act of 1981**Reasoning:** This passage describes the Economic Recovery Tax Act of 1981, an enacted fiscal measure with substantive detail about its provisions.**Input (truncated):**

The Federal Government, through tax, spending, regulatory, and monetary polices, has sacrificed long-term growth and price stability for ephemeral short term goals

Example 6: NOT_FISCAL_MEASURE**Reasoning:** This passage does not describe a specific enacted fiscal measure with substantive detail about its provisions.**Input (truncated):**

2 5 8 I 9 6 0 REPORT OF THE SiECRETARY OF THE TREASURY thereof), at the option of the owner, at any time after six months from the issue date, but only on the first day of a calendar month and upon one month's notice in writing of desire to redeem by the owner. The request for payment of the bond must be executed and certified in accordance with the provisions of the applicable regulat...

Example 7: NOT_FISCAL_MEASURE**Reasoning:** This passage does not describe a specific enacted fiscal measure with substantive detail about its provisions.**Input (truncated):**

Corporations and Other Business-Type Activities of the United States Government TABLE 108.—Comparative statement of securities of Government corporations and other business-type activities held by the Treasury, June SO, 1957-67 [Face amount, in raillions of dollars. On basis of daUy Treasury statements, see "Bases of Tables"] Agency 1957 1958 1959 1960 1961 1962 1963 1965 1966 1...

Example 8: NOT_FISCAL_MEASURE

Reasoning: This passage does not describe a specific enacted fiscal measure with substantive detail about its provisions.

Input (truncated):

NATION~L MILITARY ESTABLISHMENT-DEPARTMENT OF THE NAVY
841 PUBLIC WORKS Public Works (New), Bureau of Yards and Docks- [Public Works:] For construction, installation, and equipment of temporary or permanent public works, including public utilities and appurtenances, and necessary lands and interests therein, as authorized by the Act of June 10, 1941 (Public Law 109), as modified by the ...

Example 9: NOT_FISCAL_MEASURE

Reasoning: This passage does not describe a specific enacted fiscal measure with substantive detail about its provisions.

Input (truncated):

230 DEPARTMENT OF THE TREASURY FOCUSING ON THE NATION'S PRIORITIES—Continued TFI also works with the Office of International Affairs to direct overseas specialists who promote sound and secure financial systems in countries such as Iraq, Afghanistan, and India. In addition, TFI provides policy guidance for the Internal Revenue Service's (IRS') Criminal Investigation staff. These IRS special age...

Example 10: NOT_FISCAL_MEASURE

Reasoning: This passage does not describe a specific enacted fiscal measure with substantive detail about its provisions.

Input (truncated):

SPECIAL ANALYSES 387 Table F-7. FEDERAL ACTIVITIES IN PUBLIC WORKS (in millions of dollars) —Continued By major function and agency Function, organization unit, and program CIVIL PUBLIC WORKS-Continued Housing and Community Development—Continued District of Columbia: Loans for highway, sewage and water systems and other structures _ Grants for sewage works Federal payment to District: Grant To...

Test 12: Summary Dashboard

Test	Result	Status
T1: Chunking completeness	245/245 docs fully chunked	PASS
T2: No short chunks (min_chars filter)	0 chunks with <=100 chars (of 15061 total)	PASS

Test	Result	Status
T3: No text-less tier rows	Tier 1: 0 missing, Tier 2: 0 missing	PASS
T4: Act name alignment	Aligned→Tiers: 0 missing, Tiers→Aligned: 0 extra	PASS
T5: Passage delimiter integrity	39/39 multi-passage acts split correctly	PASS
T6: Usable passages per act	39/39 acts have ≥ 1 usable passage	PASS
T7: No within-tier duplicates	T1: 0 true dupes, 0 multi-passage chunks; T2: 0 dupes	PASS
T8: Token budget fit	Max total: 67,905 / 200,000	PASS
T9: Text encoding safety	0 unsafe chars in 270 sampled chunks	PASS
T10: Year filter consistency	Tier 1 max year: 2003, Tier 2 max year: 2007 (limit: 2007)	PASS
T11: Few-shot example preview	10 examples (5 pos, 5 neg) for fold 'Changes in Depreciation Gui...'	PASS

Overall: 11 PASS, 0 WARN, 0 FAIL out of 11 tests.

All clear: Data pipeline is validated for S2 LOOCV.